

PHPMySQL Assignment - 1

1. What is PHP? Explain its evolution and advantages.

PHP (Hypertext Preprocessor) is a server-side scripting language for web development that can be embedded in HTML.

- **Evolution:** Created in 1994, it evolved from simple tools (PHP/FI) to a robust language with the Zend Engine. PHP 7 brought major performance gains, and PHP 8 added modern features like a JIT compiler.
- **Advantages:** Open-source with a large community, platform-independent, strong database support, easy to learn, and high performance.
- **Example:**

```
<p>Hello <?php echo "World"; ?>!</p>
```

2. Define variables and constants in PHP. How are they different?

- **Variable:** A container for data whose value can change. It starts with a `$` and is case-sensitive.
 - **Example:** `$name = "John";`
- **Constant:** An identifier for a value that cannot be changed. It is defined using `define()` and has a global scope.
 - **Example:** `define("PI", 3.14);`

The key difference is that a variable's value can be reassigned, while a constant's is fixed.

3. Explain different data types available in PHP.

PHP supports several data types, including:

- **Scalar Types (single value):**
 - **Integer:** `$age = 30;` // Whole number
 - **Float:** `$price = 19.99;` // Decimal number
 - **String:** `$greeting = "Hello";` // Text
 - **Boolean:** `$is_active = true;` // true or false
- **Compound Types (multiple values):**
 - **Array:** A collection of key-value pairs.
 - **Object:** An instance of a class.
- **Special Types:**
 - **NULL:** Represents a variable with no value.

- **Resource:** A reference to an external resource.

4. What are operators in PHP? Explain any four types.

An operator is a symbol used to perform operations on variables and values.

1. **Arithmetic:** Perform math. `Example: $sum = 5 + 3; // 8`
2. **Assignment:** Assign values. `Example: $x = 10; $x += 5; // 15`
3. **Comparison:** Compare two values. `Example: if ($age == 18) { ... }`
4. **Logical:** Combine conditions. `Example: if ($is_user && $is_admin) { ... }`

5. What is a function in PHP? Differentiate call by value and call by reference.

A function is a reusable block of code that performs a specific task.

- **Call by Value (Default):** Passes a *copy* of the variable's value. The original is not affected.
 - **Example:**

```
function addFive($num) { $num += 5; }
$original = 10;
addFive($original); // $original is still 10
````*    **Call by Reference:** Passes a *reference* to the original
variable using `&`. The original is modified.
```

- **Example:**

```
function addFiveByRef(&$num) { $num += 5; }
$original = 10;
addFiveByRef($original); // $original is now 15
```

#### 6. What is an array in PHP? Explain the difference between indexed and associative arrays.

An array stores multiple values in a single variable.

- **Indexed Array:** Uses numeric keys (0, 1, 2...).
  - **Example:** `$colors = ["Red", "Green", "Blue"];`
- **Associative Array:** Uses named, string keys.
  - **Example:** `$person = ["name" => "John", "age" => 30];`

#### 7. Write short notes on looping through arrays using `each()` and `foreach()`.

- **`foreach()`:** The modern, preferred loop for iterating over arrays. Its syntax is simple and it safely works on a copy of the array.
  - **Example:**

```
foreach ($colors as $color) { echo $color; }
```

- **each()**: An older, deprecated function that is now removed from PHP. It was used with a **while** loop but has been fully replaced by **foreach**.

## 8. What are some useful library functions for array handling in PHP? Explain any five.

1. **count(\$array)**: Returns the number of elements.
  - **Example:** `echo count($colors); // Outputs 3`
2. **sort(\$array)**: Sorts an array in ascending order.
  - **Example:** `$nums = [3, 1, 2]; sort($nums); // $nums is now [1, 2, 3]`
3. **in\_array(\$val, \$arr)**: Checks if a value exists in an array.
  - **Example:** `if (in_array("Red", $colors)) { ... } // true`
4. **array\_push(\$arr, \$val)**: Adds an element to the end of an array.
  - **Example:** `array_push($colors, "Yellow");`
5. **implode(\$glue, \$arr)**: Joins array elements into a string.
  - **Example:** `$str = implode(", ", $colors); // "Red, Green, Blue"`

## 9. Explain how PHP handles HTML form data. How are **\$\_GET** and **\$\_POST** used?

PHP uses superglobal arrays to collect data submitted from HTML forms.

- **\$\_GET**: Collects data sent in the URL's query string (`method="get"`). It is visible and has a length limit, best for non-sensitive data.
  - **Example:** `$name = $_GET['username'];`
- **\$\_POST**: Collects data sent in the HTTP request body (`method="post"`). It is not visible in the URL, making it secure for sensitive data and file uploads.
  - **Example:** `$password = $_POST['user_pass'];`

## 10. What are file upload forms in PHP? Explain the steps involved and how to redirect after submission.

File upload forms allow users to send files to a server, which PHP handles via the **\$\_FILES** superglobal.

### Steps & Examples:

1. **HTML Form:** Use `method="post"` and `enctype="multipart/form-data"`.
  - **Example:** `<input type="file" name="profile_pic">`
2. **PHP Script:** Access file info via **\$\_FILES** and move it from the temporary location to a permanent one.

- **Example:** `move_uploaded_file($_FILES['profile_pic']['tmp_name'],  
"uploads/avatar.jpg");`

### Redirecting After Submission:

Use the `header()` function to prevent accidental resubmission. It must be called before any output is sent.

- **Example:**

```
// After processing...
header("Location: /thank-you.php");
exit();
```